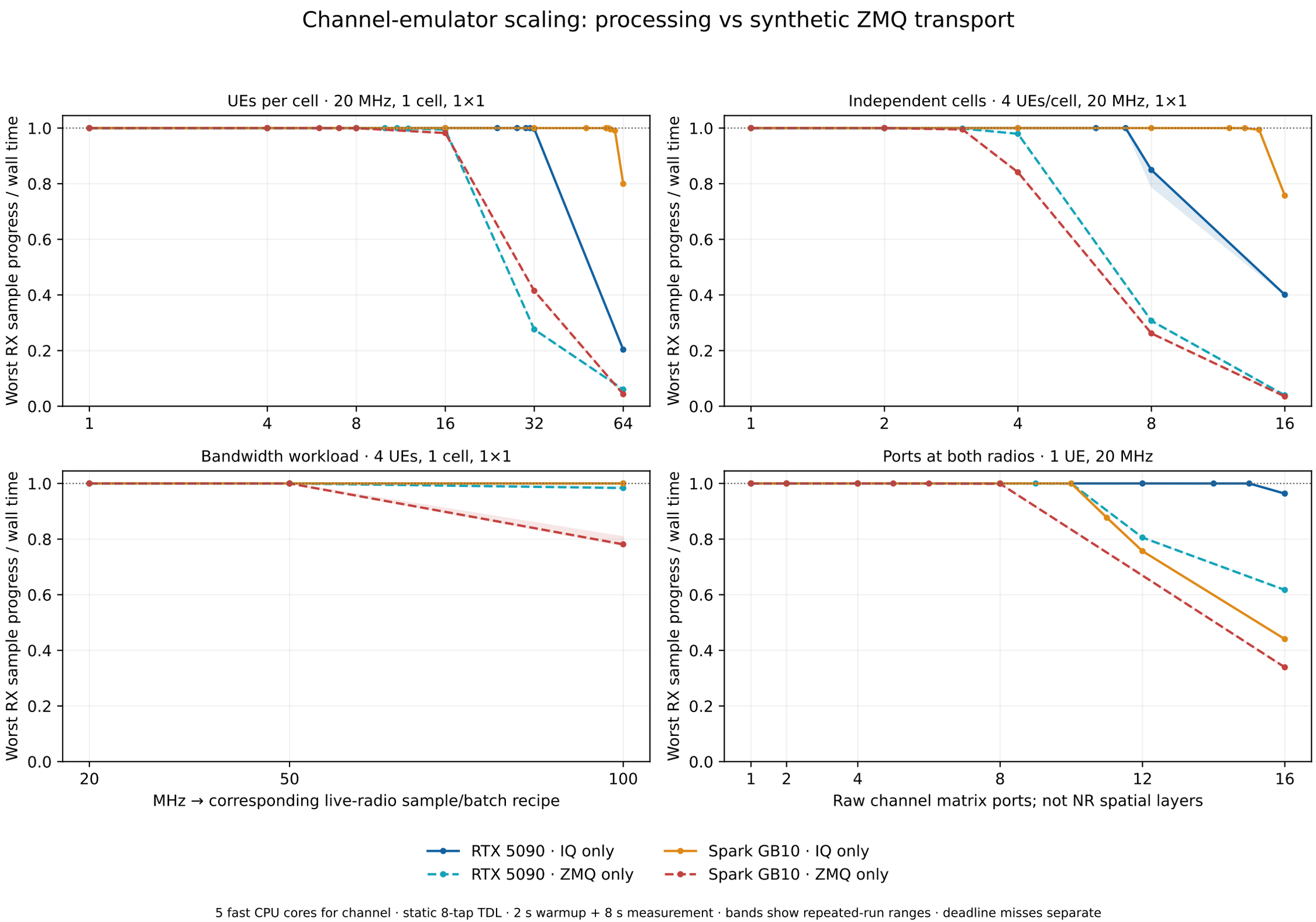
Channel-emulator scaling: processing vs synthetic ZMQ transport
UEs per cell · 20 MHz, 1 cell, 1×1
Independent cells · 4 UEs/cell, 20 MHz, 1×1
Bandwidth workload · 4 UEs, 1 cell, 1×1
Ports at both radios · 1 UE, 20 MHz
Worst RX sample progress / wall time
MHz → corresponding live-radio sample/batch recipe
Raw channel matrix ports; not NR spatial layers
RTX 5090 · IQ only
RTX 5090 · ZMQ only
Spark GB10 · IQ only
Spark GB10 · ZMQ only
5 fast CPU cores for channel · static 8-tap TDL · 2 s warmup + 8 s measurement · bands show repeated-run ranges · deadline misses separate